

Capacity Building in Biomedical Engineering Education for Digital Transformation and Industry 4.0/5.0 Technologies (BIOMED5.0)

The Overall Impact of the BIOMED5.0 Project on Biomedical Engineering Education and Industry in Pakistan

Task T5.9 - Case Study 6

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Disclaimer

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Executive Summary

This case study evaluates the perceived overall impact of the BIOMED5.0 project on biomedical engineering education and industry in Pakistan. It was jointly undertaken by the University of Engineering and Technology Taxila (UETT) and the Balochistan University of Engineering and Technology, Khuzdar (BUETK) under Task T5.9, Impact Assessment and Case Studies.

The evidence comprises seven completed Google Forms responses submitted by BIOMED5.0 focal persons or academic project staff from seven Pakistani partner HEIs. The results show a consistently positive perception of curriculum modernization, practical learning, laboratory access, digital competence, faculty development, collaboration, and innovation. All respondents reported participating in several project activities, considered the project to have had a positive overall impact in Pakistan, and reported improved digital skills at either the strongest or next-highest positive response level.

The strongest operational constraints were equipment access delays, selected by 85.7% of respondents; coordination between HEIs, 71.4%; funding shortages, 57.1%; resistance to change, 42.9%; and data collection issues, 28.6%. Communication performance was also mixed: 57.1% rated project channels very effective, while 42.9% rated them not very effective. Graduate employability remains an emerging outcome: 28.6% reported improvement, 57.1% were unsure, and 14.3% selected "not applicable".

The survey is useful for an early, cross-institutional impact snapshot and is understandable and answerable for the intended focal-person audience. However, it is not a formal test of numerical KPIs that require student, graduate, staff, or industry samples, administrative output counts, or longitudinal comparison. The report therefore links every result to the relevant grant objective, identifies whether the evidence is direct or proxy, and proposes a segmented, repeatable monitoring plan.

Priority recommendations are to strengthen equipment access, strengthen inter-HEI coordination, improve communication channels, triangulate perceptions with administrative and learning data, and repeat the survey and focus-group cycle.

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1. Introduction

1.1 Project Context and Case Study Purpose

BIOMED5.0 is an Erasmus+ Capacity Building in Higher Education project established to modernize biomedical engineering education through digital transformation and Industry 4.0/5.0 technologies. The project integrates curriculum development, laboratory modernization, VR/AR/MR teaching, a Centre of Excellence, staff and student capacity building, industry engagement, innovation and public communication.

The Grant Agreement Assigns Task T5.9 to impact assessment and case studies. Case Study 6 is defined as an assessment of the project's overall impact on biomedical engineering education and industry in Pakistan using periodic surveys and focus-group discussions. The findings are intended to support revisions, corrections, and modifications based on stakeholder feedback.

UETT and BUETK jointly developed and implemented the current survey. The case study examines whether early project effects are visible across partner institutions, where implementation is strongest, which barriers are limiting progress, and what evidence is still required before formal KPI attainment can be confirmed.

1.2 Assessment Questions

- What changes are perceived in curricula, teaching methods, practical training, laboratories and digital competence?
- What contribution is perceived in faculty development, industry-academia collaboration, the Centre of Excellence, innovation and entrepreneurship?
- What early signals are visible for student interest, employability, communication and overall impact?
- Which operational challenges require corrective action?
- How does the evidence relate to the objectives and KPIs in the Grant Agreement?

1.3 Target Respondents and Coverage

The target respondents for this survey wave were BIOMED5.0 focal persons and academic project staff with institution-level knowledge of implementation. Seven responses were received from six Pakistani partner HEIs: BUETK (two responses), Sir Syed University of Engineering and Technology, Salim Habib University, Liaquat University of Medical and Health Sciences, UET Lahore and UET Taxila. The instrument did not directly survey students, graduates, laboratory staff as a separate group, employers, or members of the general public.

1.4 Sources Used

The analysis uses the respondent-level Google Forms export, the aggregated Google Forms analytics, and the BIOMED5.0 Grant Agreement.

2. Analysis

2.1 Method and Interpretation

The questionnaire contained institution and focal-person fields followed by 20 substantive questions. It used single-choice and multiple-response items covering participation, education, facilities, competence, collaboration, societal impact, student outcomes, communication, implementation challenges and overall impact. All seven respondents completed the substantive items; no missing required responses were observed in the supplied export.

Percentages are calculated from $n=7$ and therefore move in increments of 14.3 percentage points. Results describe the perceptions of informed project participants and should be read as directional evidence. They do not represent statistically generalisable findings for all students, staff, graduates or employers, and no pre-project baseline or comparison group was supplied. The grant describes periodic surveys and focus groups; no focus-group records were included in the evidence set reviewed for this report.

2.2 Respondent Engagement

All seven respondents had participated in several BIOMED5.0 activities. Six respondents (85.7%) described themselves as very involved and one (14.3%) as moderately involved. This profile supports informed institutional commentary, while also increasing the possibility of positive-response bias because the sample consists of active project participants.

Participation and involvement in BIOMED5.0 ($n=7$)

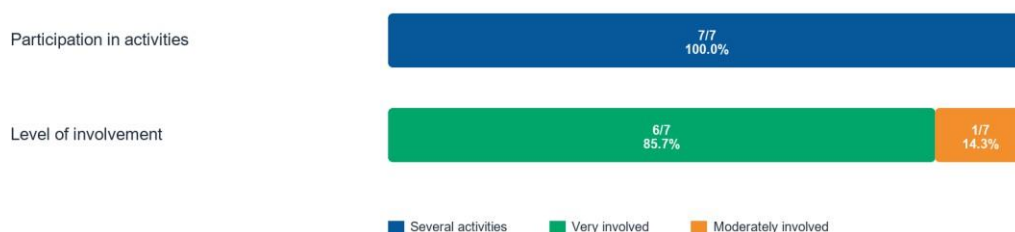


Figure 1 Participation and level of involvement among focal-person respondents.

2.3 Curriculum Modernization, Practical Learning and Laboratories

Six respondents (85.7%) reported that BIOMED5.0 had improved biomedical engineering curricula; one UETT respondent selected not applicable. Five respondents (71.4%) reported that the project introduced a lot of new hands-on training or practical skills, while two (28.6%) reported some. Five respondents observed clear improvements in teaching laboratories or equipment and two observed that facilities were somewhat better. All seven stated that student access to laboratories or equipment had been enhanced.

Digital capacity findings were also positive. Three respondents strongly agreed and four agreed that their digital skills had improved. Four observed a clear increase in digital tools, VR/AR, or Industry 4.0/5.0 technologies in teaching and learning, while three observed some increase. Five rated the project's contribution to faculty development as high and two as medium.

Perceived education, laboratory and capacity outcomes (n=7)

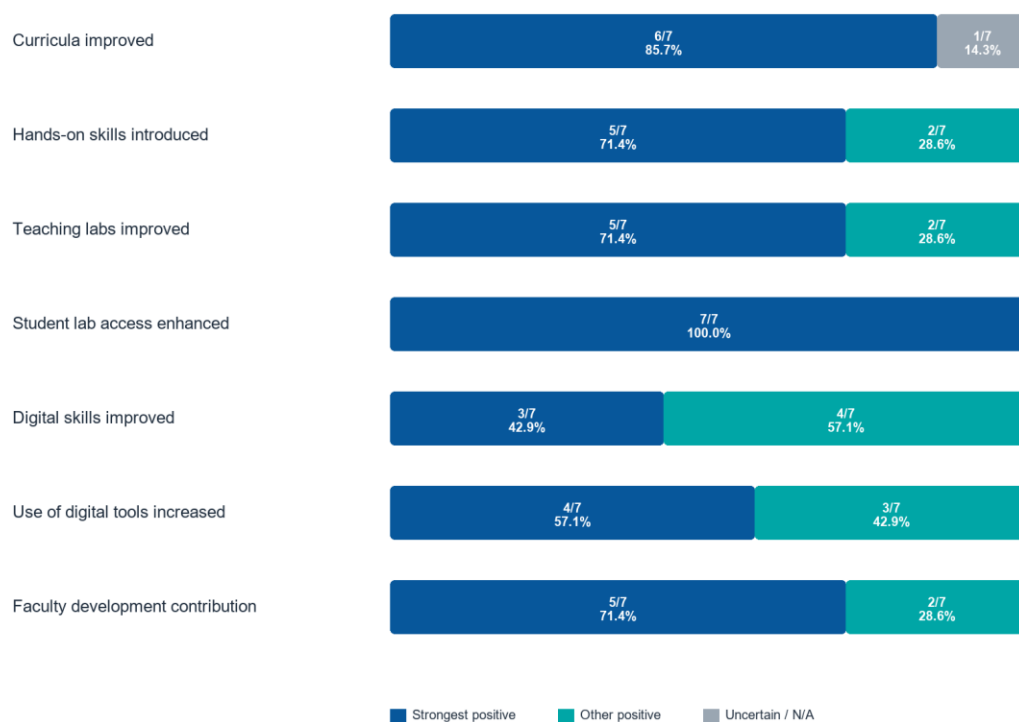


Figure 2 Perceived education, laboratory access, digital competence and faculty-development outcomes.

2.4 Changes in Teaching Methods and Course Content

Updated equipment training was the most frequently reported change, selected by six respondents (85.7%). More practical laboratories or simulations, AR/VR integration, and interdisciplinary approaches were each selected by five respondents (71.4%). Machine-learning modules were selected by four respondents (57.1%), industry case studies by three (42.9%), and online or hybrid delivery by two (28.6%).

The distribution is consistent with the project's emphasis on practical, digitally enabled and interdisciplinary teaching. Lower selection of industry case studies and online delivery indicates areas where adoption may be uneven across institutions.

Changes in teaching methods or course content (multiple response, n=7)

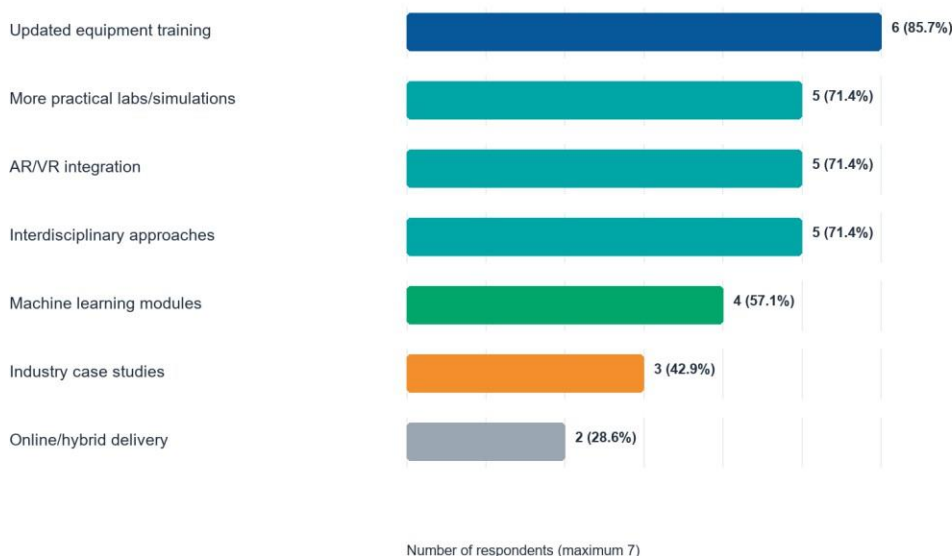


Figure 3 Reported changes in teaching methods and course content; respondents could select more than one option.

2.5 Industry Linkage, Centre of Excellence and Innovation

Five respondents (71.4%) reported that BIOMED5.0 had fostered industry-academia collaboration; one reported no, and one was unsure. The most frequently reported roles of the Societal Impact Champion or academic staff were delivery of industry training workshops (71.4%), co-development of biomedical prototypes (57.1%), and organization of focus groups or discussions (57.1%). Partnership facilitation was selected by 42.9%, mentoring by 28.6% and securing joint funding by 14.3%; policy or technology-adoption advocacy was not selected.

All respondents perceived some contribution from the Centre of Excellence to research, education and industry collaboration: 57.1% selected strongly, 28.6% somewhat and 14.3% a little. Innovation and entrepreneurship perceptions were entirely positive, with 57.1% selecting strongly and 42.9% somewhat.

Collaboration, Centre of Excellence and innovation outcomes (n=7)

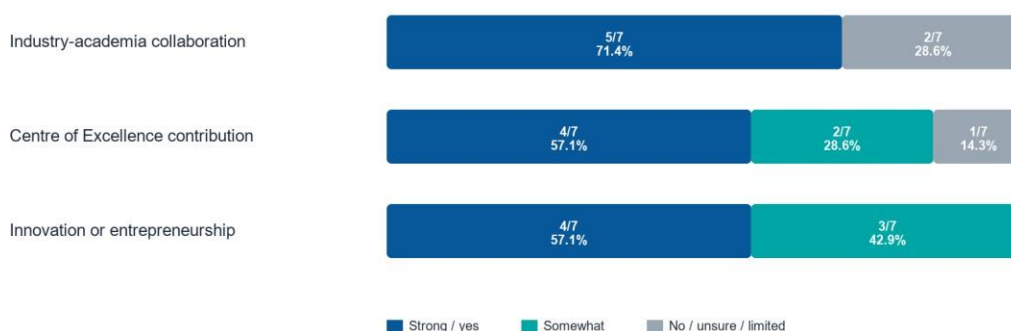


Figure 4 Perceived collaboration, Centre of Excellence and innovation outcomes.

2.6 Student Interest, Employability and Overall Impact

Five respondents (71.4%) reported growth in student enrolment or interest in biomedical engineering; one reported no growth, and one selected not applicable. Employability evidence is less mature: two respondents (28.6%) reported enhanced graduate employability, four (57.1%) were unsure and one (14.3%) selected not applicable. All seven respondents nevertheless considered BIOMED5.0 to have had a positive overall impact on biomedical engineering in Pakistan.

The contrast between the unanimous overall assessment and uncertainty about employability is important. It suggests that immediate education and capacity-building changes are more visible than labour-market outcomes, which require graduate tracking, employer feedback and a longer observation period.

Emerging student and overall outcomes (n=7)

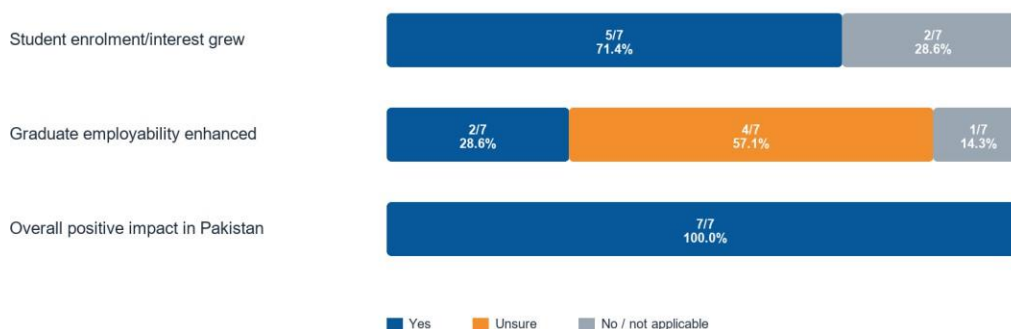


Figure 5 Early perceptions of student interest, graduate employability and overall project impact.

2.7 Communication and Implementation Constraints

Four respondents (57.1%) rated the project's website, newsletters and social-media channels as very effective, while three (42.9%) rated them not very effective. No respondent selected the intermediate 'somewhat effective' option. This split indicates unequal reach, relevance or consistency across institutions and supports targeted communication improvement rather than a single project-wide conclusion.

Although the overall impact was positive, there were a number of practical challenges that affected the implementation. The most common one was equipment-access delay reported by 85.7% of respondents. This was followed by coordination between higher education institutions (71.4%) and funding shortages (57.1%). Resistance to change was identified by 42.9%, while data-collection issues were reported by 28.6%.

These findings suggest that the effectiveness of a capacity-building project is not only dependent on the availability of funding and equipment but also on timely procurement, institutional coordination, change management, and systematizing data collection. The first three constraints should therefore receive immediate management attention.

2.8 Logical Link Between Findings and Recommendations

The survey results establish a clear progression from the identified problem to the recommended actions:

Identified gap → BIOMED5.0 intervention → observed impact → remaining challenge → recommended action

Identified Issue	Evidence from Survey	Implication	Recommended Action
Limited practical exposure	100% reported enhanced access to laboratories/equipment	Practical capacity improved	Sustain and expand modern laboratory access
Need for digital competencies	100% agreed/strongly agreed that digital skills improved	Digital transformation is producing measurable benefits	Continue digital and Industry 4.0/5.0 training
Need for modern teaching approaches	Practical labs, AR/VR, ML, equipment training, and interdisciplinary approaches were reported	Teaching practices are becoming more technology-oriented	Integrate emerging technologies permanently into curricula
Weak industry linkage	71.4% reported improved industry–academia collaboration	Project has strengthened external engagement	Expand long-term industry partnerships and joint projects
Equipment-access delays	85.7% identified equipment delays	Delays can limit timely implementation	Improve procurement, deployment, and equipment-sharing mechanisms
Institutional coordination	71.4% identified coordination as a challenge	Collaboration requires stronger mechanisms	Establish structured inter-institutional coordination
Funding limitations	57.1% identified funding shortages	Sustainability may be affected	Develop mechanisms for continued institutional and external funding
Employability impact not yet conclusive	Responses included “Yes,” “No,” and “Unsure”	Long-term employment outcomes require further evidence	Establish graduate tracking and employability monitoring

Communication effectiveness and implementation constraints (n=7)

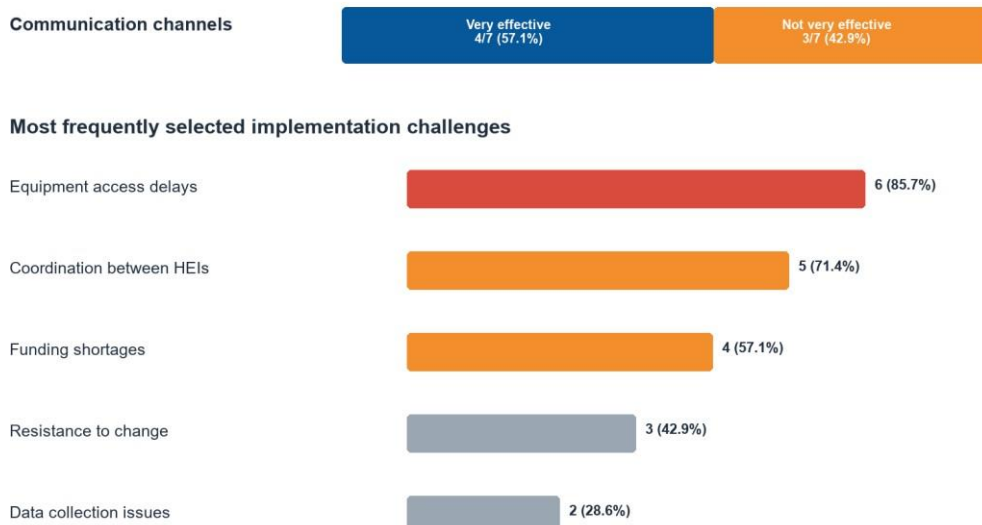


Figure 6 Communication-channel assessment and the most frequently reported implementation constraints.

2.9 Pakistan-Specific Gaps

Respondents identified four priority gaps. Shortage of AR/VR medical simulations, insufficient local industry partnerships, and resource constraints in rural HEIs were each selected by two respondents (28.6%). Limited access to advanced imaging such as MRI or CT was selected by one respondent (14.3%). No respondent selected lack of machine-learning diagnostics training, outdated curricula relative to global standards, or gender and diversity inclusion as the single most important gap. These results should be interpreted as forced priorities, not evidence that unselected gaps are absent.

2.10 Alignment with Grant Objectives and KPIs

The impact to curriculum modernization, laboratory access, VR/AR/MR teaching, digital competence, the Centre of Excellence, capacity building, collaboration, communication, and Task T5.9 case studies. The current survey addresses these themes. Its contribution is strongest as a perception-based monitoring instrument.

Table 1: Grant KPI alignment and status of evidence from this case study.

Objective / KPI	Grant requirement	Current survey evidence
SO2.2-SO2.3; medium-term student satisfaction	New/revised modules; >95% student satisfaction in digital technology learning	85.7% perceived curriculum improvement; teaching changes widely reported
SO3.1-SO3.3; T3.6-T3.8	Lab upgrades, VR/AR/MR sessions, participation and positive learning feedback	100% perceived enhanced lab access; 71.4% clear lab improvement; 71.4% AR/VR integration

Objective / KPI	Grant requirement	Current survey evidence
SO4.1-SO4.3	Functional Centre of Excellence; training, collaboration, innovation and entrepreneurship	100% perceived some CoE contribution; 71.4% industry collaboration; 100% positive innovation perception
SO5.2; T5.4	Awareness, adoption and stakeholder engagement through the Societal Impact Champion	100% overall positive impact; communication rated very effective by 57.1% and not very effective by 42.9%
T5.9; D5.1 project assessment	Impact case studies produced, published and used for feedback	This report documents Case Study 6 and identifies corrective actions
Medium-term staff digital competence	95% satisfaction in digital competency achieved through BIOMED5.0 activities via T5.9	100% agreed/strongly agreed digital skills improved; 100% rated faculty development high/medium

3 Recommendation and Implementation Plan

3.10 Priority Recommendations

Strengthen coordination between HEIs. Use a monthly implementation forum, a common action register and a concise evidence dashboard tied to work-package outputs.

Improve communication effectiveness. Segment messages by stakeholder group and HEI; establish a predictable publishing calendar; monitor reach, engagement and feedback rather than publication volume alone.

Convert the survey into a measurement system. Maintain the focal-person instrument as an institutional pulse survey, while adding respondent-specific instruments for students, staff, graduates, employers and event participants.

Triangulate perceptions with verifiable records. Link survey results to approved curricula, module delivery, laboratory inventories, attendance, assessment results, CoE activity, partnerships, prototypes, start-ups and communication analytics.

Measure employability longitudinally. Establish baseline and follow-up graduate destination data and employer feedback before drawing conclusions on the grant's employability targets.

Close the feedback loop. Document decisions, corrective actions and subsequent changes after every survey or focus group, as required by the Case Study 6 description.

3.11 Resources and Risk Controls

The plan primarily requires staff time, a shared evidence repository, basic survey and dashboard tools, data-protection controls, and designated institutional evidence owners. The main risks are low response rates among direct beneficiaries, incompatible data definitions, delayed administrative records, and social-desirability bias. Mitigations include short role-specific instruments, an agreed KPI dictionary, a minimum evidence set, anonymous collection where suitable, reminder schedules, and transparent reporting of missing data.

4 Conclusion

The case study provides a clear early signal that BIOMED5.0 is positively influencing biomedical engineering education across the participating institutions. Respondents reported strong engagement, curriculum modernisation, practical and laboratory improvements, increased digital competence, faculty development, collaboration, Centre of Excellence activity, innovation and growth in student interest. The unanimous positive overall assessment is supported by multiple domain-specific results rather than a single general question.

The evidence also identifies practical weaknesses that could reduce project impact if left unresolved. Equipment access delays, inter-HEI coordination, and funding are the leading constraints. Communication is uneven, and employability outcomes are not yet mature enough for a confident conclusion.

In response to the project coordinator's observations, this report explicitly identifies the target respondents, evaluates the questionnaire's usefulness and answerability, maps result to relevant objectives and KPIs, and separates proxy evidence from formal attainment. Implementing the proposed measurement and action plan will allow future survey waves and focus groups to provide stronger, respondent-appropriate and verifiable evidence for Task T5.9 and the final project assessment.

5 Exhibits

5.10 Respondent Composition

Table 2 Participating institutions and number of focal-person responses.

Institution	Responses	Share
The Balochistan University of Engineering and Technology, Khuzdar (BUETK)	2	28.6%
Sir Syed University of Engineering and Technology	1	14.3%
Salim Habib University	1	14.3%
Liaquat University of Medical and Health Sciences	1	14.3%
University of Engineering and Technology Lahore	1	14.3%
University of Engineering and Technology Taxila (UETT)	1	14.3%

5.11 Detailed Survey Results

Table 3 Distribution of substantive survey responses.

Question / indicator	Response categories reported	Distribution
Participation in BIOMED5.0 activities	Several	7 (100%)
Level of involvement	Very involved; moderately involved	6 (85.7%); 1 (14.3%)
Curricula improved	Yes; not applicable	6 (85.7%); 1 (14.3%)
Hands-on training/practical skills	A lot; some	5 (71.4%); 2 (28.6%)
Teaching laboratory/equipment change	Clear improvement; somewhat better	5 (71.4%); 2 (28.6%)
Student access to labs/equipment	Yes	7 (100%)
Digital skills improved	Strongly agree; agree	3 (42.9%); 4 (57.1%)
Use of digital tools/VR/AR increased	Clearly; somewhat	4 (57.1%); 3 (42.9%)
Faculty-development contribution	High; medium	5 (71.4%); 2 (28.6%)
Industry-academia collaboration	Yes; no; unsure	5 (71.4%); 1 (14.3%); 1 (14.3%)
Centre of Excellence contribution	Strongly; somewhat; a little	4 (57.1%); 2 (28.6%); 1 (14.3%)
Innovation/entrepreneurship promoted	Strongly; somewhat	4 (57.1%); 3 (42.9%)
Student enrolment/interest grew	Yes; no; not applicable	5 (71.4%); 1 (14.3%); 1 (14.3%)
Graduate employability enhanced	Yes; unsure; not applicable	2 (28.6%); 4 (57.1%); 1 (14.3%)
Communication channels effective	Very effective; not very effective	4 (57.1%); 3 (42.9%)
Overall positive impact in Pakistan	Yes	7 (100%)

5.12 Multiple-Response Items

Table 4 Counts for multiple-response teaching, industry-role and implementation items.

Item	Options in descending count	Counts
Teaching changes	Updated equipment training; practical labs/simulations; AR/VR; interdisciplinary; ML; industry cases; online/hybrid	6; 5; 5; 5; 4; 3; 2
Societal Impact Champion/academic roles	Industry training; prototypes; focus groups; partnerships; mentoring; joint funding; policy advocacy	5; 4; 4; 3; 2; 1; 0
Implementation challenges	Equipment delay; coordination; funding; resistance; data collection; staff training; infrastructure	6; 5; 4; 3; 2; 0; 0
Pakistan-specific priority gap	AR/VR simulations; local industry links; rural resources; advanced imaging	2; 2; 2; 1

5.13 Evidence Limitations

- 5.13.1 Small purposive sample: seven focal-person or academic project respondents from six HEIs.
- 5.13.2 BUETK contributed two responses; other represented HEIs contributed one each.
- 5.13.3 No direct student, graduate, laboratory-staff, employer, or general-public sample was included.
- 5.13.4 No baseline, comparison group, focus-group record, or inferential statistical analysis was available.
- 5.13.5 Several questions are broad perception measures and are not numerically equivalent to the Grant Agreement's KPIs.
- 5.13.6 Percentages should always be read together with counts because one response equals 14.3 percentage points.

6 References

- [1] European Commission. Grant Agreement - Project 101129077, BIOMED5.0. Relevant sections: project objectives and KPIs; intervention logic; short-, medium- and long-term impacts; WP5; Task T5.9; Deliverable D5.1.
- [2] BIOMED5.0 Impact Assessment Case Study Survey: The Overall Impact of the BIOMED5.0 Project on Biomedical Engineering. Google Forms aggregate analytics, exported 13 August 2026, 7 responses.
- [3] BIOMED5.0 Impact Assessment Case Study Survey: All University Responses. Google Forms respondent-level export, exported 13 August 2026, 64 pages.